Encoded PowerShell Execution Detection and Investigation using Sysmon and Wazuh

Overview

This report documents the investigation and detection workflow for Base64-encoded PowerShell execution activity identified using Sysmon and Wazuh within a controlled SOC detection engineering lab environment.

The simulation focused on identifying attacker-style PowerShell execution techniques commonly associated with:

- fileless malware
- obfuscated PowerShell scripts
- malicious PowerShell loaders
- defense evasion behavior
- staged attack execution

The investigation demonstrates how Sysmon telemetry and Wazuh correlation rules can be used together to:

- detect suspicious PowerShell behavior
- analyze process execution chains
- investigate encoded command execution
- validate alerts
- reduce false positives
- perform structured SOC-style investigations

This report is part of the larger SOC Detection Engineering Lab project implemented using Sysmon and Wazuh. The master report already documents the overall lab architecture, telemetry pipeline, and attack simulation environment.

Objective

The objective of this simulation was to investigate how Sysmon and Wazuh detect Base64-encoded PowerShell execution techniques commonly associated with fileless malware, obfuscated scripts, and attacker PowerShell tradecraft.

MITRE ATT&CK Mapping

Tactic	Technique	ID
Execution	PowerShell	T1059.001
Defense Evasion	Obfuscated/Encoded Files and Information	T1027

Attack Simulation

The following PowerShell command was executed in the Windows endpoint lab environment:

```
powershell.exe -nop -w hidden -ep bypass -EncodedCommand VwByAGkAdABlAC0ATwB1AHQAcAB1AHQAI
AAiAEgAZQBsAGwAbwAiAA==
```

Command Breakdown

Parameter	Description
-nop	Starts PowerShell without loading profile

Parameter	Description
-w hidden	Hides PowerShell window
-ep bypass	Bypasses PowerShell execution policy
-EncodedCommand	Executes Base64-encoded command

Sysmon Investigation

Key Sysmon Events Identified

Sysmon Event ID	Detection Purpose
1	Process Creation
11	File Creation

These events provided the primary endpoint telemetry for the investigation and aligned with the PowerShell monitoring and detection coverage implemented in the SOC lab environment.

Sysmon Event ID 1 — Process Creation Analysis

Key Findings

Field	Value
Event ID	1
Process	powershell.exe
Parent Process	powershell.exe
User	socadmin
Integrity Level	High
Suspicious Flags	-nop -w hidden -ep bypass -EncodedCommand

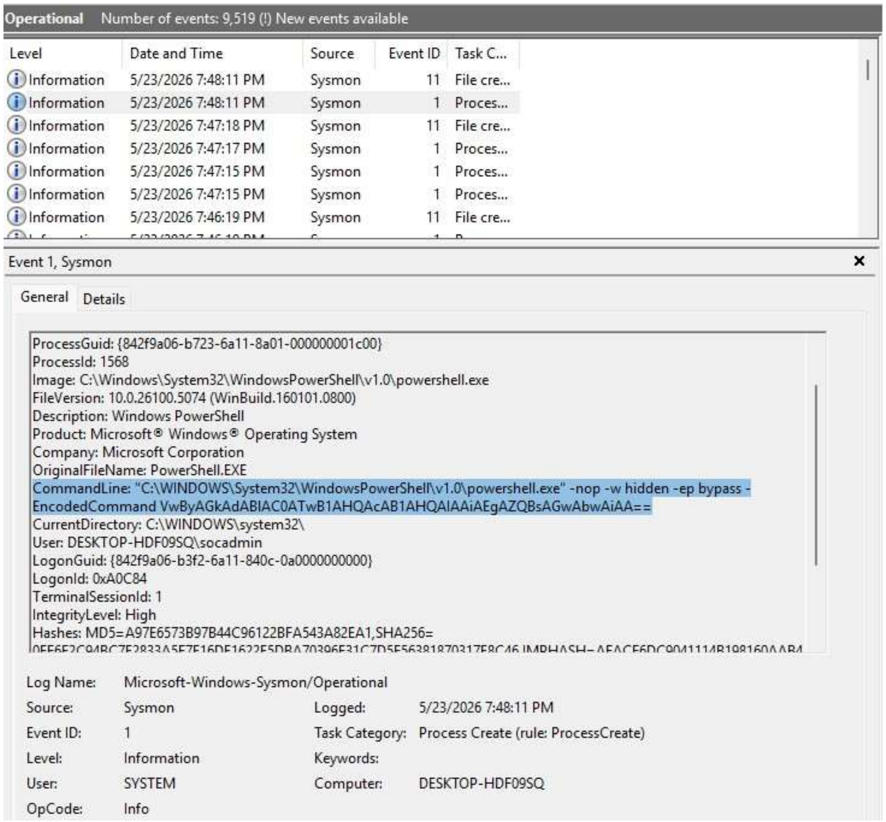
Investigation Analysis

Sysmon Event ID 1 captured the execution of a PowerShell process containing multiple suspicious command-line arguments associated with attacker tradecraft and defense evasion techniques.

The process executed with:

- hidden window execution
- execution policy bypass
- Base64-encoded command execution
- elevated privileges

The parent-child PowerShell relationship further increased suspicion because nested PowerShell execution is commonly observed in malicious PowerShell loaders and staged attack activity.



Encoded Payload Analysis

Base64 Payload

Encoded Command


```
VwByAGkAdABlAC0ATwB1AHQAcAB1AHQAIAAiAEgAZQBsAGwAbwAiAA==
```


Decoded PowerShell

```
Write-Output "Hello"
```

Investigation Analysis

Although the decoded payload was benign, the execution behavior itself remained highly suspicious.

Detection confidence was based on:

- encoded command execution
- hidden PowerShell execution
- execution policy bypass
- elevated PowerShell context
- nested PowerShell process creation

This demonstrates the importance of behavioral detection over payload content alone.

File Creation Analysis

Sysmon Event ID 11 — File Creation

Sysmon Event ID	Detection
11	File Creation

Artifact Observed

```
__PSScriptPolicyTest_xxx.ps1
```

File Location

```
C:\Users\socadmin\AppData\Local\Temp\
```

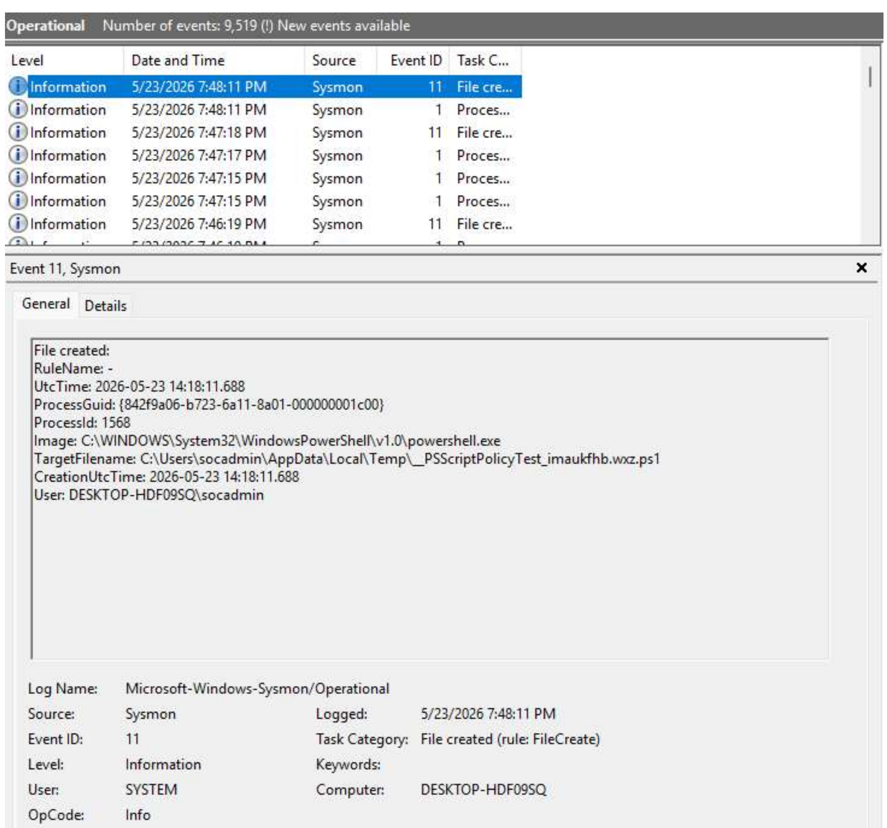
Investigation Analysis

Sysmon Event ID 11 captured the creation of a temporary PowerShell script policy test file within the user Temp directory.

This artifact is generated internally by PowerShell during execution policy validation and is commonly observed during PowerShell execution.

Although the file itself was not malicious, its creation provided additional telemetry supporting the overall PowerShell execution timeline.

This investigation step was important because it demonstrated false-positive awareness during SOC analysis and helped differentiate legitimate PowerShell behavior from malicious file-dropping activity.



Wazuh Detection Analysis

Wazuh Rules Triggered

Rule ID	Severity	Description
92057	12	Encoded PowerShell execution detected
92213	15	File created in malware-associated directory

Wazuh Alert Investigation

Wazuh successfully detected the encoded PowerShell execution using behavioral correlation rules designed to identify Base64-encoded PowerShell activity.

The detection logic identified:

- PowerShell spawning PowerShell
- use of the -EncodedCommand parameter
- suspicious execution arguments
- elevated execution context

The alert was correctly mapped to MITRE ATT&CK technique T1059.001 (PowerShell).

False Positive Validation

The temporary PowerShell policy test file triggered an additional Wazuh alert related to suspicious file creation activity.

However, investigation confirmed that the file was a legitimate PowerShell execution artifact rather than a malicious payload.

This highlights the importance of analyst validation and false-positive reduction during SOC investigations.

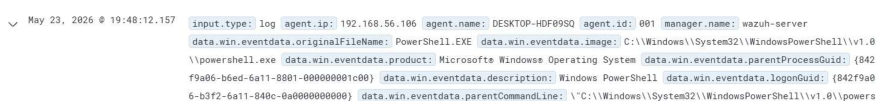


Figure 3 – Wazuh detection for Base64-encoded PowerShell execution

Detailed Wazuh alert telemetry for encoded PowerShell activity :


```
{
"_index": "wazuh-alerts-4.x-2026.05.23",
"_id": "vBozVZ4BdId3fDzgfGvQ",
"_version": 1,
"_score": null,
"_source": {
"input": {
"type": "log"
},
"agent": {
"ip": "192.168.56.106",
"name": "DESKTOP-HDF09SQ",
"id": "001"
},
"manager": {
"name": "wazuh-server"
},
"data": {
```

"win": {
"eventdata": {
"originalFileName": "PowerShell.EXE",
"image": "C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe",
"product": "Microsoft® Windows® Operating System",
"parentProcessGuid": "{842f9a06-b6ed-6a11-8801-000000001c00)",
"description": "Windows PowerShell",
"logonGuid": "{842f9a06-b3f2-6a11-840c-0a0000000000)",
"parentCommandLine": "\\\"C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\\\"",
"processGuid": "{842f9a06-b723-6a11-8a01-000000001c00)",
"logonId": "0xa0c84",
"parentProcessId": "10052",
"processId": "1568",
"currentDirectory": "C:\\\\WINDOWS\\\\system32\\\\",
"utcTime": "2026-05-23 14:18:11.195",
"hashes":
"MD5=A97E6573B97B44C96122BFA543A82EA1,SHA256=0FF6F2C94BC7E2833A5F7E16DE1622E5DBA70396F31C7D5
"parentImage": "C:\\\\Windows\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe",
"company": "Microsoft Corporation",
"commandLine": "\\\"C:\\\\WINDOWS\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe\\\" -nop -w hidden -ep bypass -EncodedCommand VwByAGkAdABlAC0ATwB1AHQAcAB1AHQAIAAiAEgAZQBsAGwAbwAiAA==",
"integrityLevel": "High",
"fileVersion": "10.0.26100.5074 (WinBuild.160101.0800)",
"user": "DESKTOP-HDF09SQ\\\\socadmin",
"terminalSessionId": "1",
"parentUser": "DESKTOP-HDF09SQ\\\\socadmin"
},
"system": {
"eventID": "1",
"keywords": "0x8000000000000000",
"providerGuid": "{5770385f-c22a-43e0-bf4c-06f5698ffbd9)",
"level": "4",
"channel": "Microsoft-Windows-Sysmon/Operational",
"opcode": "0",
"message": "\"Process Create:\r\nRuleName: -\r\nUtcTime: 2026-05-23 14:18:11.195\r\nProcessGuid: {842f9a06-b723-6a11-8a01-000000001c00)\r\nProcessId: 1568\r\nImage: C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe\r\nFileVersion: 10.0.26100.5074 (WinBuild.160101.0800)\r\nDescription: Windows PowerShell\r\nProduct: Microsoft® Windows® Operating System\r\nCompany: Microsoft Corporation\r\nOriginalFileName: PowerShell.EXE\r\nCommandLine: \"C:\\WINDOWS\\System32\\WindowsPowerShell\\v1.0\\powershell.exe\" -nop -w hidden -ep bypass -EncodedCommand VwByAGkAdABlAC0ATwB1AHQAcAB1AHQAIAAiAEgAZQBsAGwAbwAiAA==\r\nCurrentDirectory: C:\\WINDOWS\\system32\\\r\nUser: DESKTOP-HDF09SQ\\socadmin\r\nLogonGuid: {842f9a06-b3f2-6a11-840c-0a0000000000)\r\nLogonId: 0xA0C84\r\nTerminalSessionId: 1\r\nIntegrityLevel: High\r\nHashes: MD5=A97E6573B97B44C96122BFA543A82EA1,SHA256=0FF6F2C94BC7E2833A5F7E16DE1622E5DBA70396F31C7D5F
{842f9a06-b6ed-6a11-8801-000000001c00)\r\nParentProcessId: 10052\r\nParentImage: C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe\r\nParentCommandLine: \"C:\\Windows\\System32\\WindowsPowerShell\\v1.0\\powershell.exe\" \r\nParentUser: DESKTOP-HDF09SQ\\socadmin\"",
"version": "5",
"systemTime": "2026-05-23T14:18:11.5489298Z",
"eventRecordID": "14809",
"threadID": "4376",
"computer": "DESKTOP-HDF09SQ",
"task": "1",
"processID": "3596",
"severityValue": "INFORMATION",
"providerName": "Microsoft-Windows-Sysmon"
}

}
},
"rule": {
"firedtimes": 3,
"mail": true,
"level": 12,
"description": "Powershell.exe spawned a powershell process which executed a base64 encoded command",
"groups": [
"sysmon",
"sysmon_eid1_detections",
"windows"
],
"mitre": {
"technique": [
"PowerShell"
],
"id": [
"T1059.001"
],
"tactic": [
"Execution"
]
},
"id": "92057"
},
"location": "EventChannel",
"decoder": {
"name": "windows_eventchannel"
},
"id": "1779545892.2885597",
"timestamp": "2026-05-23T14:18:12.157+0000"
},
"fields": {
"timestamp": [
"2026-05-23T14:18:12.157Z"
]
},
"sort": [
1779545892157
]
}

Figure 4 – Wazuh detection for Executable file dropped in folder commonly used by malware

{
"_index": "wazuh-alerts-4.x-2026.05.23",
"_id": "vRozVZ4BdId3fDzgfGvR",
"_version": 1,
"_score": null,
"_source": {
"input": {
"type": "log"

```
  },
  "agent": {
    "ip": "192.168.56.106",
    "name": "DESKTOP-HDF09SQ",
    "id": "001"
  },
  "manager": {
    "name": "wazuh-server"
  },
  "data": {
    "win": {
      "eventdata": {
        "image": "C:\\\\WINDOWS\\\\System32\\\\WindowsPowerShell\\\\v1.0\\\\powershell.exe",
        "processGuid": "{842f9a06-b723-6a11-8a01-000000001c00}",
        "processId": "1568",
        "utcTime": "2026-05-23 14:18:11.688",
        "targetFilename": "C:\\\\Users\\\\socadmin\\\\AppData\\\\Local\\\\Temp\\\\__PSScriptPolicyTest_imaukfhb.wxz.ps1",
        "creationUtcTime": "2026-05-23 14:18:11.688",
        "user": "DESKTOP-HDF09SQ\\\\socadmin"
      },
      "system": {
        "eventID": "11",
        "keywords": "0x8000000000000000",
        "providerGuid": "{5770385f-c22a-43e0-bf4c-06f5698ffbd9}",
        "level": "4",
        "channel": "Microsoft-Windows-Sysmon/Operational",
        "opcode": "0",
        "message": "\"File created:\r\nRuleName: -\r\nUtcTime: 2026-05-23 14:18:11.688\r\nProcessGuid: {842f9a06-
b723-6a11-8a01-000000001c00}\r\nProcessId: 1568\r\nImage:
C:\\WINDOWS\\System32\\WindowsPowerShell\\v1.0\\powershell.exe\r\nTargetFilename:
C:\\Users\\socadmin\\AppData\\Local\\Temp\\__PSScriptPolicyTest_imaukfhb.wxz.ps1\r\nCreationUtcTime: 2026-05-23
14:18:11.688\r\nUser: DESKTOP-HDF09SQ\\socadmin\"",
        "version": "2",
        "systemTime": "2026-05-23T14:18:11.7226624Z",
        "eventRecordID": "14810",
        "threadID": "4376",
        "computer": "DESKTOP-HDF09SQ",
        "task": "11",
        "processID": "3596",
        "severityValue": "INFORMATION",
        "providerName": "Microsoft-Windows-Sysmon"
      }
    }
  },
  "rule": {
    "firedtimes": 6,
    "mail": true,
    "level": 15,
```

```
    "description": "Executable file dropped in folder commonly used by malware",
    "groups": [
      "sysmon",
      "sysmon_eid11_detections",
      "windows"
    ],
    "mitre": {
      "technique": [
        "Ingress Tool Transfer"
      ],
      "id": [
        "T1105"
      ],
      "tactic": [
        "Command and Control"
      ]
    },
    "id": "92213"
  },
  "location": "EventChannel",
  "decoder": {
    "name": "windows_eventchannel"
  },
  "id": "1779545892.2892075",
  "timestamp": "2026-05-23T14:18:12.162+0000"
 },
 "fields": {
  "timestamp": [
    "2026-05-23T14:18:12.162Z"
  ]
 },
 "sort": [
  1779545892162
 ]
}
```

Attack Timeline

Time	Event
19:48:9	Encoded PowerShell process launched
19:48:11	Sysmon Event ID 1 generated
19:48:11	Sysmon Event ID 11 generated
19:48:11	Temporary PowerShell script file created
19:48:12	Wazuh generated encoded PowerShell detection
19:48:12	Wazuh generated executable file dropped in folder commonly used by malware

Key Detection Findings

Finding	Impact
Encoded PowerShell detected	High-confidence suspicious behavior
Finding	Impact
Execution policy bypass observed	Defense evasion indicator
Hidden PowerShell execution	Increased attack suspicion
Nested PowerShell process chain	Behavioral correlation improvement
Elevated PowerShell context	Increased severity
Temp PS policy file observed	Additional telemetry artifact

Sysmon Event IDs

Sysmon Event ID	Meaning
1	Process Creation
11	File Creation

Wazuh Rule IDs

Rule ID	Purpose
92057	Encoded PowerShell execution
92213	Suspicious file creation

Investigation Outcome

The investigation successfully demonstrated how encoded PowerShell activity can be detected using behavioral telemetry collected from Sysmon and correlated through Wazuh detection rules.

The combination of:

- command-line analysis
- parent-child process relationships
- execution context
- PowerShell behavior
- file creation telemetry

significantly improved detection confidence and investigation quality.

The investigation also reinforced the importance of:

- behavioral correlation
- contextual analysis
- false-positive validation
- PowerShell monitoring
- structured SOC investigation methodology

Conclusion

This simulation demonstrated how Sysmon and Wazuh provide strong visibility into encoded PowerShell execution techniques commonly associated with attacker tradecraft and fileless malware activity.Behavioral correlation between PowerShell execution, command-line arguments, parent-child process relationships, and file creation telemetry significantly improved detection confidence.

The investigation also highlighted the importance of analyst validation and contextual analysis when distinguishing legitimate PowerShell artifacts from malicious behavior.